

AJAY KUMAR

Data Engineer

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SUMMARY

Data Engineer with over five years of experience building scalable cloud data pipelines and analytics platforms. Led projects such as migrating healthcare data to AWS, improving reporting and ETL performance, and optimizing workflows for data reliability. Skilled in AWS, Snowflake, Python, and SQL, regularly partnering with cross-functional teams to support analytics and AI/ML needs. Aiming to use hands-on data engineering expertise to help deliver robust and efficient data solutions.

TECHNICAL SKILLS

- **Cloud Platforms:** Amazon Web Services (EC2, S3, Glue, Redshift, Lambda, EMR), Snowflake
- **Data Engineering:** ETL development, data modeling, dbt, data warehousing, data ingestion, data validation, workflow orchestration
- **Programming:** Python (Pandas, NumPy, PySpark), SQL, Shell Scripting, Git
- **Databases & Storage:** Snowflake, Redshift, PostgreSQL, MySQL, Hive, Data Lake / Data Warehouse
- **Visualization:** Power BI, Tableau, Excel (Pivot Tables)
- **Streaming & Processing:** Apache Spark, Kafka, Airflow, Apache Beam
- **Compliance & Domain Knowledge:** FHIR, HL7, HIPAA compliance, Healthcare data integration
- **Methodologies:** Agile/Scrum, CI/CD Automation, GitHub Actions, JIRA

PROFESSIONAL EXPERIENCE

Humana | Data Engineer | United States | January 2024 – Present

- Engineered a cloud-based healthcare data platform on AWS by migrating on-prem claims and member datasets, improving reporting performance by 35% and enabling scalable enterprise analytics.
- Architected and optimized scalable ETL pipelines for large-scale FHIR and HL7 datasets using AWS Glue and PySpark, reducing end-to-end processing time by 40% and improving data reliability.
- Implemented Kafka and Spark jobs to process large-scale streaming device data, enabling low-latency analytics and improving decision-making speed.
- Automated CI/CD-driven ETL deployments and workflow orchestration using Airflow, dbt, and GitHub Actions, reducing manual intervention by 50% and improving pipeline reliability.
- Operationalized production-ready datasets for AI/ML workloads using dbt and Airflow, improving model reliability, minimizing data inconsistencies, and accelerating development cycles
- Built and scheduled dbt models in Snowflake (incremental loads, marts), adding dbt tests to catch data issues before downstream reporting.
- Ingested multi-format data (APIs, CSV, JSON, Parquet) into AWS S3 and Snowflake, engineering production-grade datasets that powered analytics and reporting for 4+ business units.
- Wrote SQL and Python validation checks that identified missing, duplicate, and malformed records before entering reporting layers, resulting in higher data quality and fewer downstream errors.

Cognizant | Data Engineer | India | February 2021 – December 2022

- Enhanced large-scale Spark and EMR jobs to handle high-volume batch processing, cutting runtime by 35% and enhancing pipeline reliability.
- Orchestrated Kafka pipelines to enable near real-time data availability instead of relying only on overnight batch jobs.
- Designed ETL workflows using Glue and Airflow, reducing manual data preparation effort and improving data consistency across sources.
- Built and maintained AWS-based pipelines to centralize data from diagnostic and healthcare systems, delivering reporting used by multiple business teams.
- Partnered with data science teams to deliver production-ready datasets and resolve upstream data issues, accelerating model development.
- Created Power BI and Tableau dashboards used by 10+ stakeholders to monitor device utilization and performance trends.
- Implemented Python and SQL scripts to clean incoming data, resolving recurring data quality issues across multiple source systems.

Synechron | Data Analyst | India | January 2020 – February 2021

- Developed automated Python and SQL pipelines to process millions of financial records, enhancing reporting accuracy and reducing manual effort.
- Designed interactive Tableau, Power BI, and Excel dashboards to deliver actionable insights for 15+ business users, enabling informed business strategies.

- Resolved performance bottlenecks in SQL queries and ETL logic, cutting report refresh times from many hours to under 45 minutes.
- Partnered with a team of 4+ analysts and stakeholders to validate datasets and perform UAT, which ensured reports met business requirements and reduced rework
- Streamlined ad hoc financial analyses and reporting tasks using Python, SQL, and Excel, increasing team productivity and reducing error rates by 25%.

EDUCATION

Master of Science in Data Science | January 2023 – December 2024

University of Houston-Clear Lake, Houston, Texas, USA

Bachelor of Technology in Computer Science and Engineering | June 2017 – May 2021

Gandhi Institute of Technology and Management, Visakhapatnam

CERTIFICATIONS

- Completed Python for Everybody – Coursera, University of Michigan.
- Achieved IBM Data Science certification – Coursera, IBM.